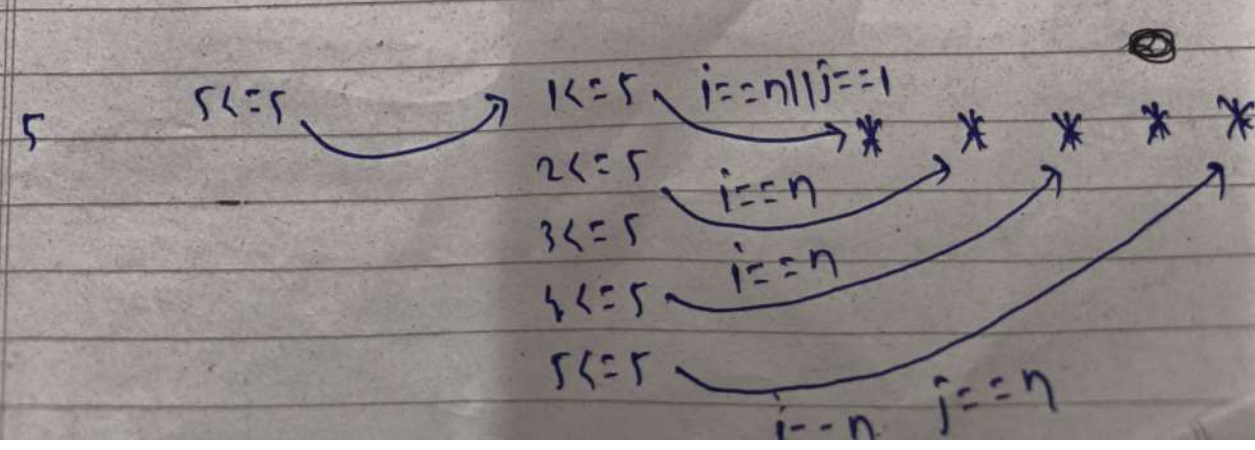
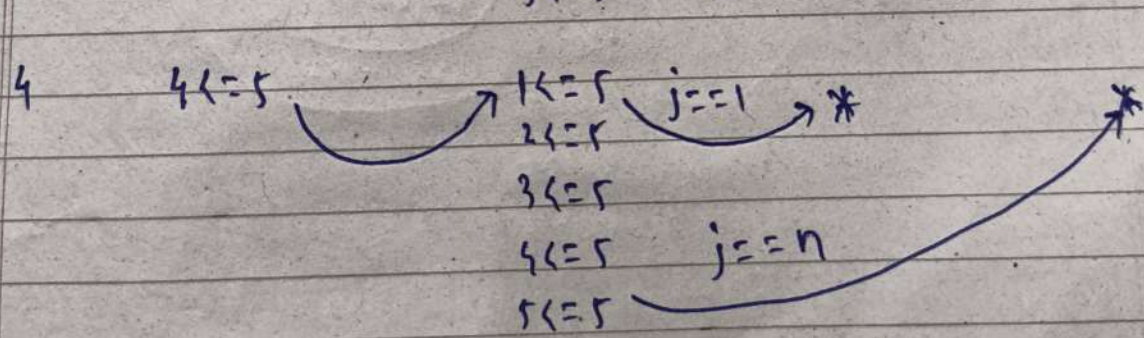
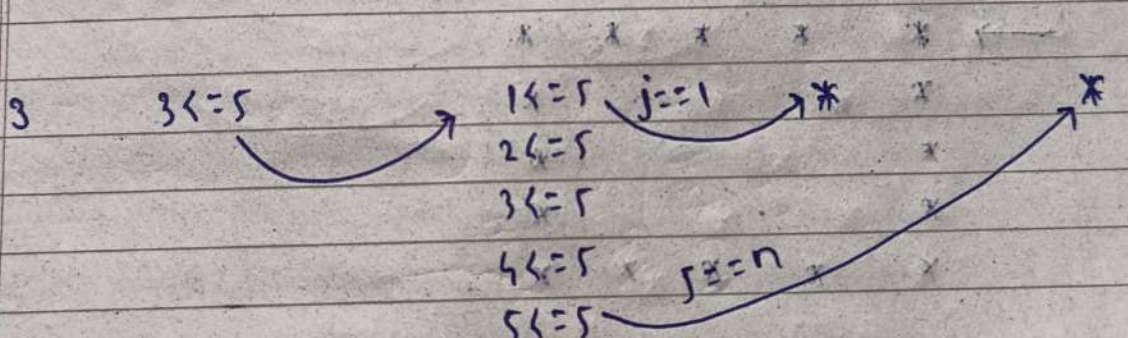
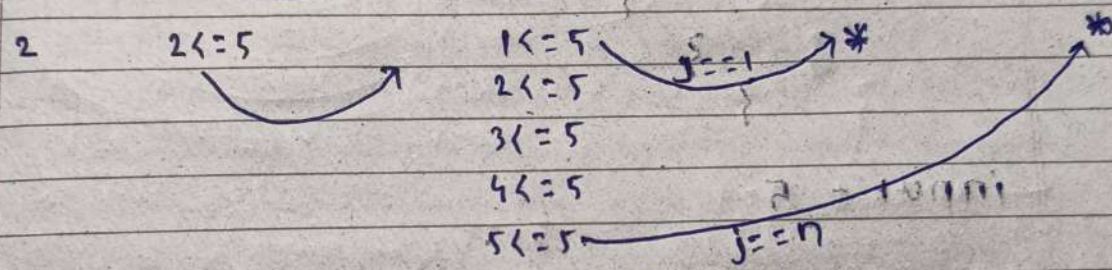
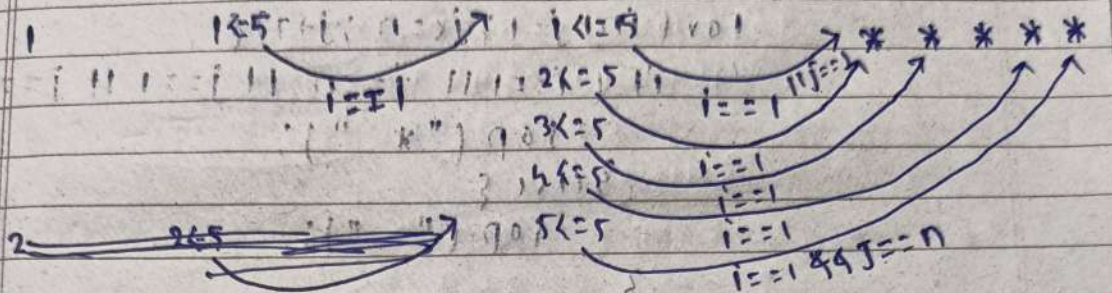


① Hollow Square
n = 5

int i = 1; i <= n; i++
{
int j = 1; j <= n; j++
{
if (i == 1 || i == n || j == 1 || j == n)
cout << "X" << " ";
else
cout << " ";
}}




```

import java.util.*;
class HollowSquare {
    public static void main (String args[]) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        for (int i=0; i<n; i++) {
            for (int j=0; j<n; j++) {
                if (i==0 || i==n-1 || j==0 || j==n-1) {
                    System.out.print("* ");
                } else {
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
    }
}

```

input - 5

```

→ *   *   *   *   *
 *   *   *   *   *
 *   *   *   *   *
 *   *   *   *   *
 *   *   *   *   *

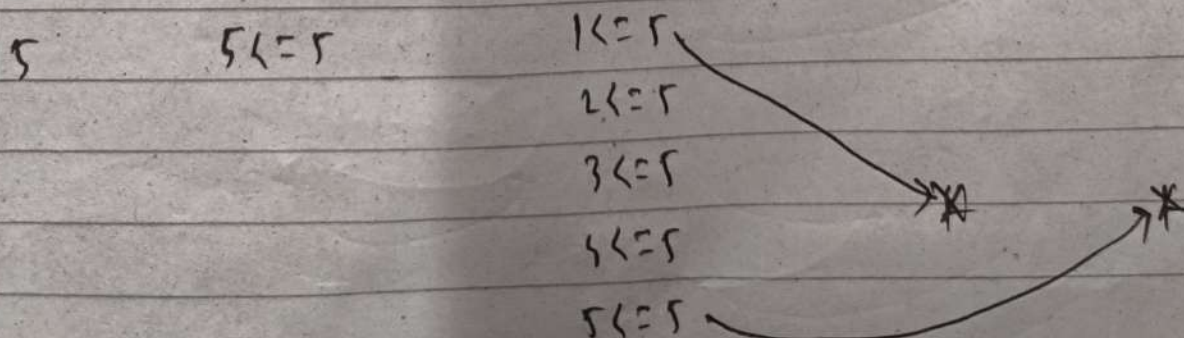
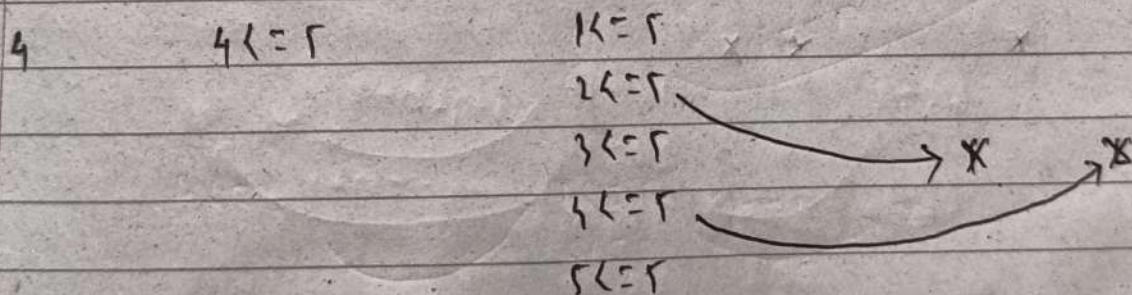
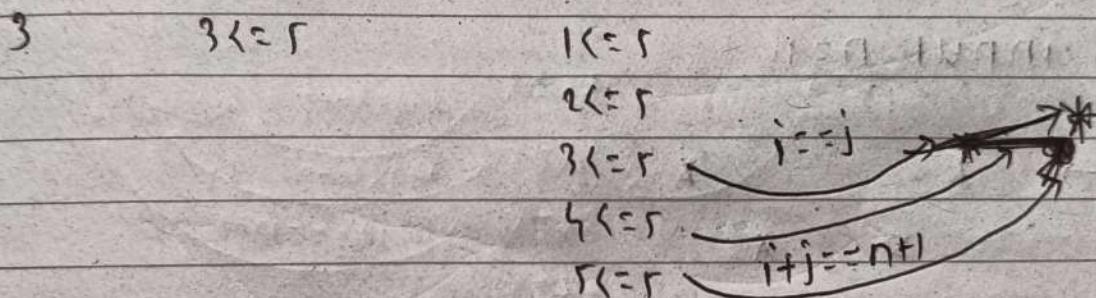
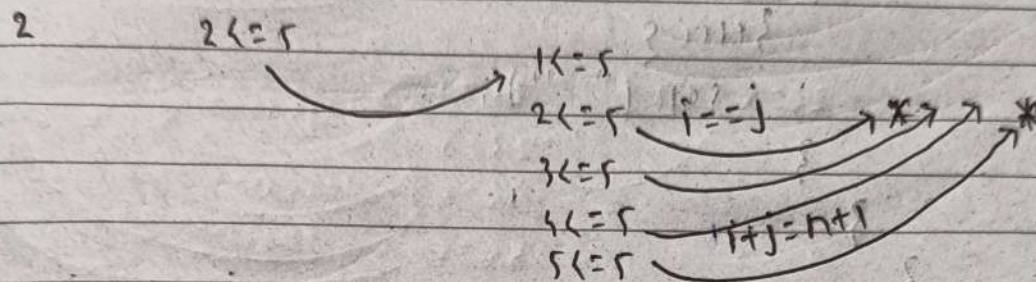
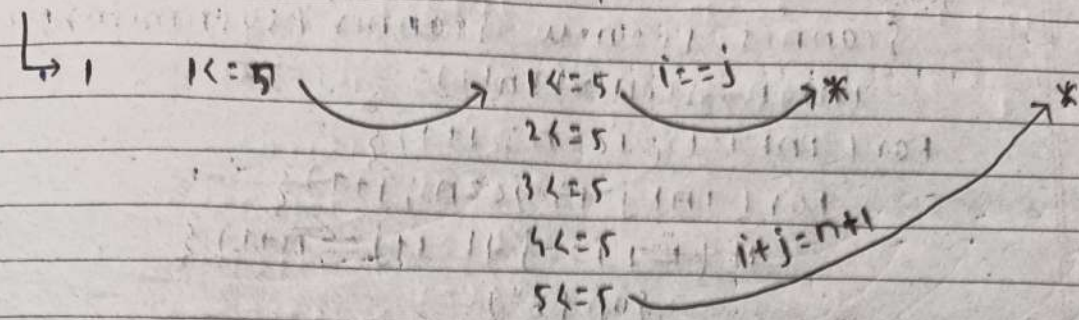
```


②

Diagonal square

input $n=5$

int $i=1$ $k=n$




```
import java.util.*;
```

```
class X {
```

```
    public static void main () {
```

```
        Scanner sc = new Scanner (System.in);
```

```
        int n = sc.nextInt();
```

```
        for (int i=1; i<=n; i++) {
```

```
            for (int j=1; j<=n; j++) {
```

```
                if (i==j i==j || i+j==n+1) {
```

```
                    sep("*")
```

```
                } else {
```

```
                    sep(" ");
```

```
            }
```

```
        }
```

```
    }
```

— input n = 5

→

```

*           *
 *         *
  *       *
   *     *
  *     *
 *     * *
*     * * *
```


③

Right triangle

$n = 5$

int i = 1

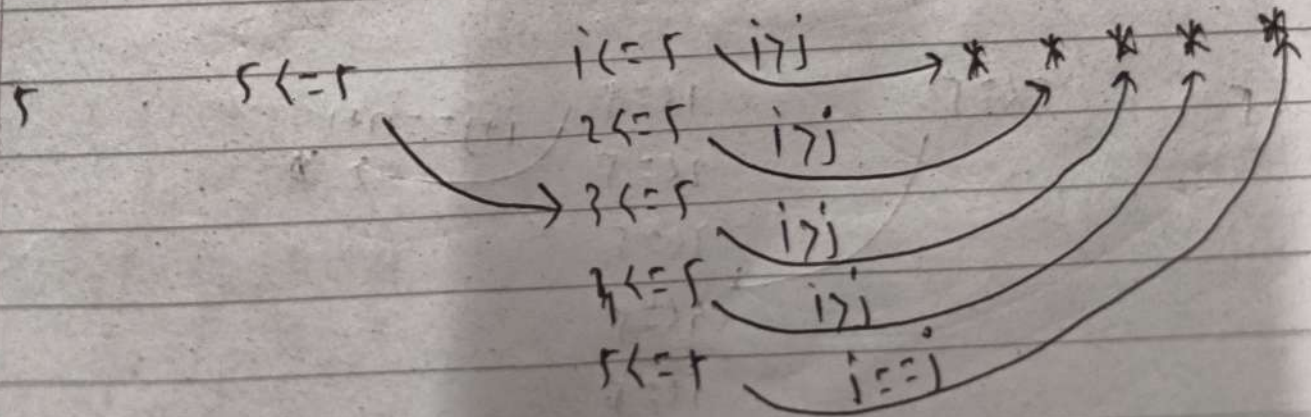
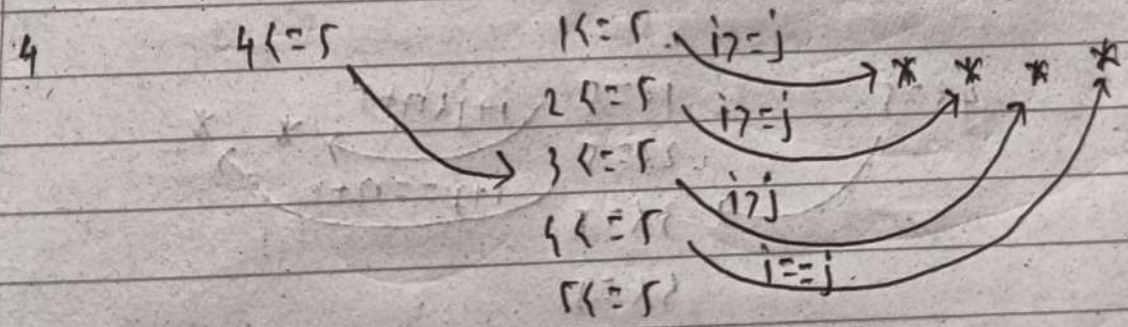
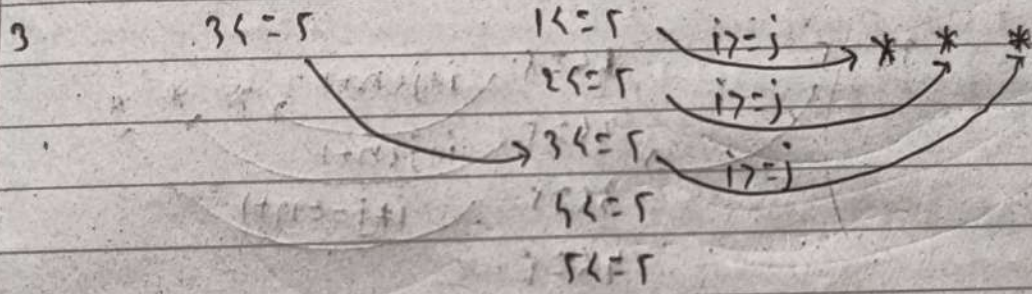
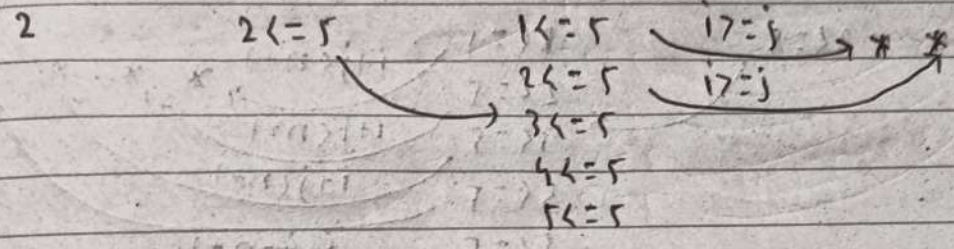
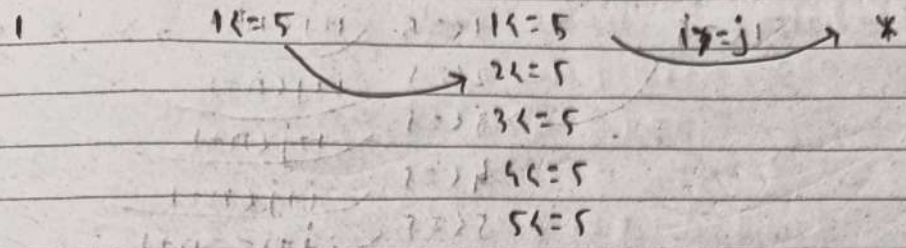
ik = n

int j = 1

* x *

x x x x

x x x x x



④

Inverted Right Angle →

$n=5$

$\text{int } i=1$

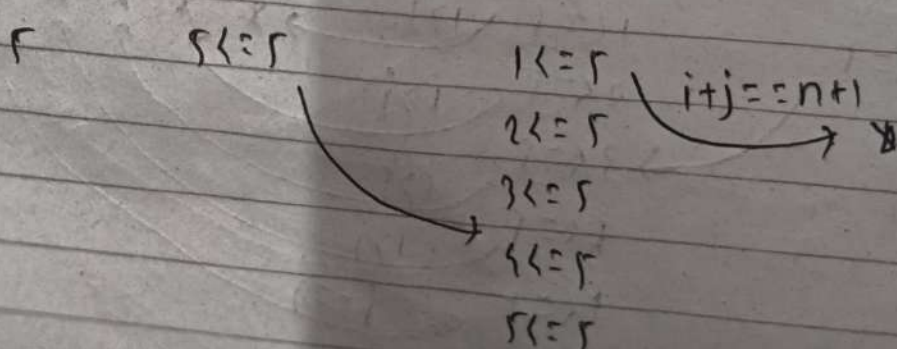
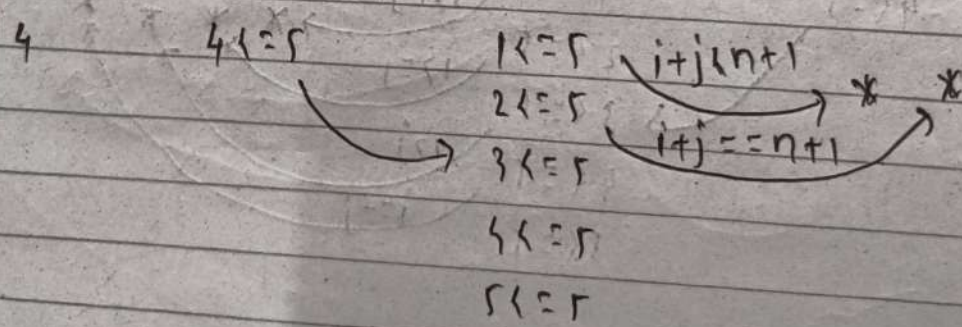
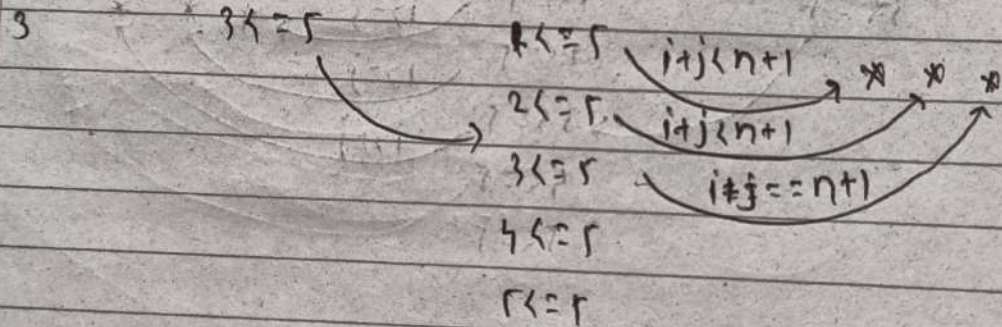
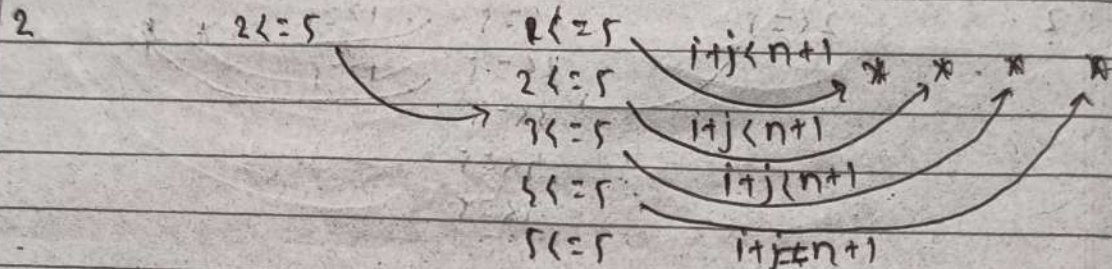
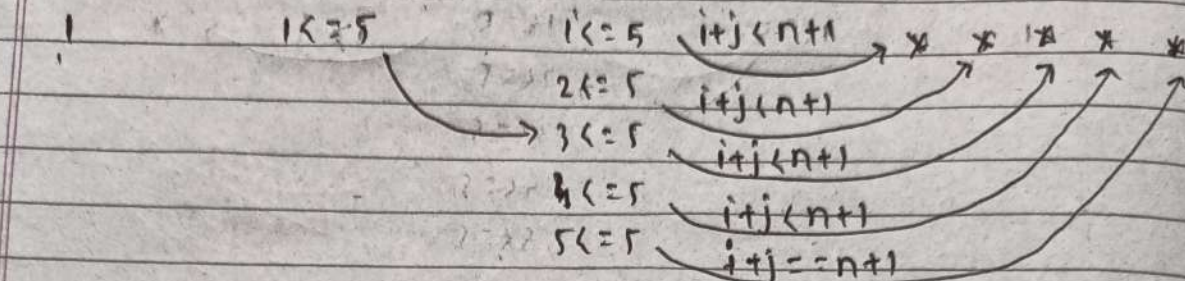
$i \leq n$

$\text{int } j=1$

```

x x x x x
x x x x
x x x
x x
x

```



5

Chess Board Pattern

→

* # * #

* # *

* # * #

* # *

n = 4

int i = 1

i <= n

int j = 1

1

i <= 4

1 <= 4

$i+j \% 2 == 0$

* # * #

2 <= 4

$i+j \% 2 == 0$

3 <= 4

$i+j \% 2 == 0$

4 <= 4

$i+j \% 2 == 0$

2

i <= 4

1 <= 4

$i+j \% 2 == 0$

2 <= 4

$i+j \% 2 == 0$

3 <= 4

$i+j \% 2 == 0$

4 <= 4

$i+j \% 2 == 0$

3

i <= 4

1 <= 4

$i+j \% 2 == 0$

2 <= 4

$i+j \% 2 == 0$

3 <= 4

$i+j \% 2 == 0$

4 <= 4

$i+j \% 2 == 0$

4

i <= 4

1 <= 4

$i+j \% 2 == 0$

2 <= 4

$i+j \% 2 == 0$

3 <= 4

$i+j \% 2 == 0$

4 <= 4

$i+j \% 2 == 0$

input n = 4

→

* # * #

* # *

* # * #

* # *